

TAKAMATSU, Japan

WMO: 478910

Lat: 34.3183N

Lon: 134.0542E

Elev: 11

StdP: 101.20

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.3	0.6	-8.7	1.8	3.8	-7.4	2.0	4.5	9.5	6.7	8.7	6.1	2.3	220	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	34.6	25.3	33.6	25.1	32.5	25.0	26.4	31.8	26.0	31.3	25.6	30.7	3.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.0	20.1	28.9	24.6	19.6	28.7	24.1	19.0	28.6	82.3	31.9	80.5	31.3	78.8	30.8	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.9	6.8	5.9	-2.0	36.3	1.0	0.8	-2.7	36.9	-3.3	37.4	-3.8	37.8	-4.5	38.4
			-3.9	27.1	1.1	0.5	-4.7	27.5	-5.3	27.8	-5.9	28.1	-6.7	28.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.9	5.7	6.2	9.5	14.8	20.0	23.6	27.9	29.0	25.0	19.3	13.2	8.1	(6)
(7)		DBStd	8.45	2.05	2.45	2.89	3.11	2.35	2.31	2.23	1.85	2.74	2.82	2.99	2.53	(7)
(8)		HDD10.0	366	133	109	46	3	0	0	0	0	0	5	69	(8)	
(9)		HDD18.3	1618	391	339	275	109	11	0	0	0	22	155	317	(9)	
(10)		CDD10.0	2888	1	3	29	148	309	407	554	589	448	287	102	11	(10)
(11)		CDD18.3	1100	0	0	0	5	62	157	296	331	198	50	1	0	(11)
(12)		CDH23.3	10423	0	0	0	30	311	964	3264	4051	1626	178	1	0	(12)
(13)		CDH26.7	3992	0	0	0	3	42	234	1363	1851	480	21	0	0	(13)
(14)	Wind	WSAvg	2.5	2.9	2.7	2.7	2.5	2.4	2.2	2.3	2.4	2.3	2.1	2.2	2.7	(14)
(15)	Precipitation	PrecAvg	1159	39	45	80	71	105	155	154	112	172	120	57	47	(15)
(16)		PrecMax	1621	99	87	199	169	301	349	351	433	528	417	142	111	(16)
(17)		PrecMin	769	2	7	32	29	11	22	14	12	29	20	14	4	(17)
(18)		PrecStd	265	23	22	38	29	61	85	95	98	126	98	32	27	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	16.6	20.7	26.3	29.7	33.0	35.5	36.1	33.8	28.8	22.4	17.0	(19)
(20)			MCWB	9.1	11.5	12.4	15.3	18.3	24.1	25.3	25.5	25.1	22.3	17.0	13.1	(20)
(21)		2%	DB	12.2	13.9	18.2	23.8	27.8	30.7	34.3	35.0	32.1	26.7	20.9	15.4	(21)
(22)			MCWB	7.7	9.4	11.5	15.0	17.5	22.5	25.1	25.5	24.5	20.7	16.2	11.4	(22)
(23)		5%	DB	10.9	12.4	16.6	22.2	26.4	29.1	33.3	34.0	30.7	25.2	19.7	14.1	(23)
(24)			MCWB	6.8	8.2	10.8	14.4	17.2	21.5	25.1	25.2	23.8	19.5	15.5	10.1	(24)
(25)		10%	DB	9.8	10.8	14.8	20.5	25.1	27.8	32.2	33.0	29.4	24.0	18.4	12.8	(25)
(26)			MCWB	6.1	6.7	9.7	13.7	16.9	21.0	25.0	25.1	23.1	18.7	14.6	9.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	13.1	14.9	19.3	21.9	25.1	26.8	27.1	26.1	23.8	19.0	14.6	(27)
(28)			MCD B	12.8	15.5	17.8	22.0	25.2	30.4	32.5	32.7	31.2	27.1	20.5	16.0	(28)
(29)		2%	WB	8.7	10.7	13.2	17.3	20.5	24.2	26.2	26.6	25.4	22.2	17.6	12.3	(29)
(30)			MCD B	10.9	13.0	16.1	20.8	23.8	28.6	31.7	32.1	30.1	25.5	19.7	14.4	(30)
(31)		5%	WB	7.6	8.9	11.9	16.1	19.4	23.5	25.8	26.2	24.9	20.9	16.4	10.9	(31)
(32)			MCD B	10.2	11.2	15.1	19.9	23.1	27.4	31.1	31.5	29.2	23.9	18.9	13.2	(32)
(33)		10%	WB	6.6	7.5	10.5	15.0	18.5	22.6	25.4	25.8	24.3	19.7	15.1	9.6	(33)
(34)			MCD B	9.4	10.2	14.2	18.9	22.6	26.1	30.5	30.9	28.1	22.9	17.9	12.2	(34)
(35)	Mean Daily Temperature Range	MDBR	6.8	7.5	8.4	9.1	8.8	7.0	6.9	7.2	6.8	7.3	7.5	7.1	(35)	
(36)		5% DB	MCD BR	8.5	9.6	11.0	11.6	11.1	8.9	8.3	8.3	7.9	8.2	8.3	8.6	(36)
(37)			MCWBR	5.8	6.2	6.5	5.3	4.5	3.2	2.5	2.6	3.0	4.1	5.2	5.9	(37)
(38)		5% WB	MCD BR	8.0	8.1	9.0	8.8	8.0	7.1	7.6	7.5	6.8	6.5	7.0	7.5	(38)
(39)			MCWBR	6.2	6.2	6.6	5.3	4.6	3.0	2.6	2.7	2.9	4.1	5.1	6.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.399	0.445	0.505	0.535	0.539	0.558	0.558	0.531	0.491	0.449	0.420	0.393	(40)	
(41)		taud	2.166	2.058	1.913	1.892	1.952	1.980	2.032	2.096	2.178	2.210	2.196	2.199	(41)	
(42)		Ebn at Noon	780	783	767	767	769	752	749	762	775	773	753	760	(42)	
(43)		Edh at Noon	118	146	183	196	187	181	171	158	139	124	114	108	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.48	3.07	4.04	4.87	5.27	4.71	5.10	5.25	4.09	3.38	2.57	2.19	(44)	
(45)		RadStd	0.17	0.24	0.29	0.45	0.50	0.45	0.65	0.51	0.39	0.34	0.22	0.14	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page